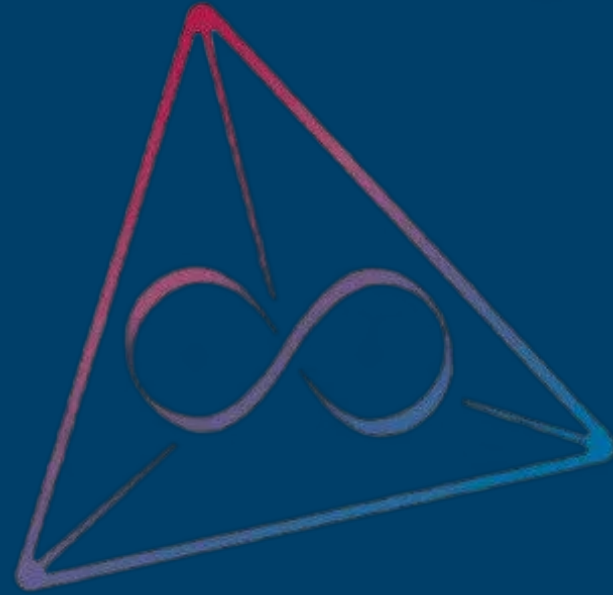


Bier spheres in algebra, geometry and combinatorics

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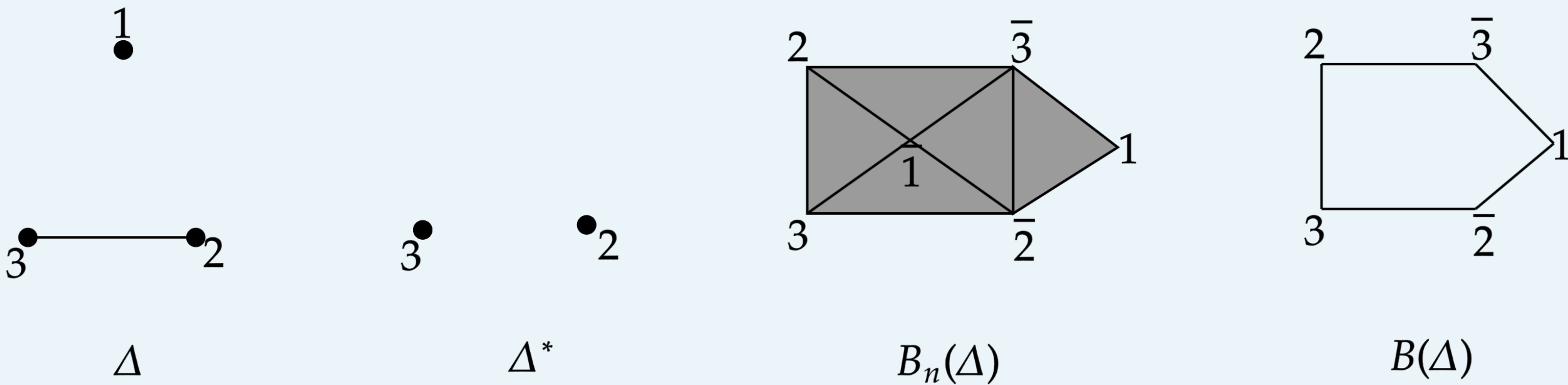
Summary

- Bier spheres were introduced by Bier in 1992 as a construction that takes an arbitrary simplicial complex Δ , and outputs a simplicial sphere $B(\Delta)$. A very similar construction had already been considered 2 years earlier by Villarreal in commutative algebra, where the goal was to find large classes of Cohen-Macaulay quadratic squarefree monomial ideals. In this poster we go through known and conjectured properties of this construction in algebra, geometry and combinatorics.
- In algebra, we use this construction to produce h -vectors of Gorenstein artinian rings that satisfy extreme properties.
 - In geometry, we use this construction to produce thousands of new examples of centrally symmetric simplicial spheres that are polytopal, but cannot be realized as the boundary of a centrally symmetric polytope.
 - In graph theory, a conjecture of Levit and Mandrescu says that certain graphs have a unimodal independence polynomial. These independence polynomials are the f -vectors of certain simplicial balls whose boundary is a Bier sphere.

Definitions of Bier spheres and balls

- There are at least two equivalent ways to describe the Bier sphere $B(\Delta)$ of a simplicial complex Δ .
- (Bier 1992) Let Δ be a simplicial complex on vertex set $[n]$. The **Bier sphere** $B(\Delta)$ of Δ is the deleted join of Δ and its Alexander dual Δ^* . Bier showed that $B(\Delta)$ is a simplicial sphere of dimension $n - 2$.
 - (Villarreal 1990) Let Δ be a simplicial complex on vertex set $[n]$. The **Bier ball** $B_n(\Delta)$ of Δ is the simplicial complex on vertex set $[n] \sqcup [\bar{n}]$ whose minimal nonfaces are:
 1. $\tau \subseteq [n]$, where τ is a minimal nonface of Δ ;
 2. $\{i, \bar{i}\}$ for $i \in [n]$.If we add the minimal nonfaces $\bar{\tau}$ for τ a minimal nonface of Δ^* , we obtain the Bier sphere $B(\Delta)$.

Example



The minimal nonfaces of Δ are 12, 13, while the minimal nonfaces of $B_n(\Delta)$ are 12, 13, $1\bar{1}$, $2\bar{2}$, $3\bar{3}$, and the extra minimal nonfaces of $B(\Delta)$ are $\bar{1}$ and $\bar{2}\bar{3}$. The extra nonfaces are exactly the minimal nonfaces of Δ^* on the set $[\bar{n}]$.

The algebraic story: when is a vector the h -vector of a Gorenstein ring?

- Does every h -vector of a Gorenstein algebra satisfy the conditions from the g -theorem? In particular, is every h -vector of a Gorenstein algebra unimodal?
- Stanley showed in 1978 that the answer is no by producing a Gorenstein algebra with h -vector
- (1, 13, 12, 13, 1).

His construction was an example of a **Nagata idealization**. In 1995 Boij strengthned this result by producing Gorenstein algebras with arbitrarily many peaks in their h -vectors.

In 2021, Nagata idealizations were again used by D’Alì and Venturello to find counterexamples to the *algebraic Gal’s conjecture*. Their construction was based on Bier spheres.

A Roller Coaster theorem for Gorenstein algebras

Theorem.(H-Nicklasson, 2026) Let π be a permutation of $\{1, \dots, \lfloor \frac{e}{2} \rfloor\}$, where e is a positive integer. Then there exists a Gorenstein artinian algebra A with socle degree e and h -vector $(h_0, h_1, \dots, h_e)$ such that

$h_{\pi(1)} < h_{\pi(2)} < \dots < h_{\pi(\lfloor e/2 \rfloor)}$

The proof is **not constructive**, and instead uses techniques from graph theory, and the corresponding Roller Coaster theorem for independence polynomials of graphs.

What does a “Roller Coaster algebra” look like?

Let $S_{\Delta} = \frac{R}{(x_1^2, \dots, x_n^2) + I_{\Delta}}$ where Δ is pure of dimension $d - 1$. The algebras are of the form

$A_{\Delta} = \left\{ \begin{bmatrix} f & g \\ 0 & f \end{bmatrix} : f \in S \text{ and } g \in \omega_{S_{\Delta}} \right\}.$

Here, $\omega_{S_{\Delta}}$ is (after shifting) the canonical module of S_{Δ} . In this case, we can think of $\omega_{S_{\Delta}}$ as being generated by monomials $m_1, \dots, m_k$ whose support is a facet of Δ .

The h -vector of A_{Δ} is given by

$h_i = f_{i-1}(\Delta) + f_{d-i+1}(\Delta).$

In particular, A is isomorphic to the Nagata idealization of the Stanley-Reisner ring of $B_n(\Delta)$

The geometric story: where is the hay in the haystack?

- In 1984, Bokowski, Ewald and Kleinschmidt found the first example of a polytopal simplicial sphere that admits a combinatorial symmetry that cannot be realized geometrically by any of its realizations. Spheres with nonrealizable symmetries are very hard to construct. Until 2026, this was the only known explicit example of such a sphere.
- In 2005, Björner et al. showed that most Bier spheres are not polytopal. To this day, **there is no known example of a nonpolytopal Bier sphere**. In 2025 Timotijević et al. showed every Bier sphere up to 11 vertices is polytopal.
- $B(\Delta)$ is centrally symmetric (cs) if and only if Δ is self-dual. Can we find nonpolytopal cs Bier spheres? Are there cs Bier spheres that cannot be realized by a cs polytope?

Simplicial polytopes with nonrealizable (central) symmetries

Theorem. (H-Yang, 2026+) There exist cs Bier spheres that are polytopal but not cs-polytopal. In particular, these spheres have noncontractible realization spaces. Moreover, they can be taken to be cs k -neighborly for several values of k .

There are at least 2098 such spheres. Their facets are known explicitly.

- The proof of noncs-polytopality uses a very explicit criterion, together with a cs version of Gale diagrams.
- The proof of polytopality uses an *algorithm written by Claude Opus 4.7*.

A simple criterion for noncs-polytopality

- Let n be even and Δ a self-dual simplicial complex on $[n]$. Assume the following:
1. every minimal nonface of Δ has at least $k + 1 \geq 3$ elements
 2. There exists distinct $r, \ell \in [n]$ and distinct $C, D \subseteq [n] \setminus \{r, \ell\}$ with $|C| = |D| = \frac{n-2}{2}$ such that:
 - 2.1 $C \cup r \in \Delta$ and $[n] \setminus (C \cup \ell) \in \Delta$
 - 2.2 $D \cup \ell \in \Delta$ and $[n] \setminus (D \cup r) \in \Delta$
- Then $B(\Delta)$ is not cs-polytopal.

Example.
 $\Delta = \langle 123, [6] \setminus 124 = 356, 246, [6] \setminus 236 = 145, 125, 126, 134, 136, 146, 156, \rangle$
($r = 3, \ell = 4, C = 12, D = 26$)

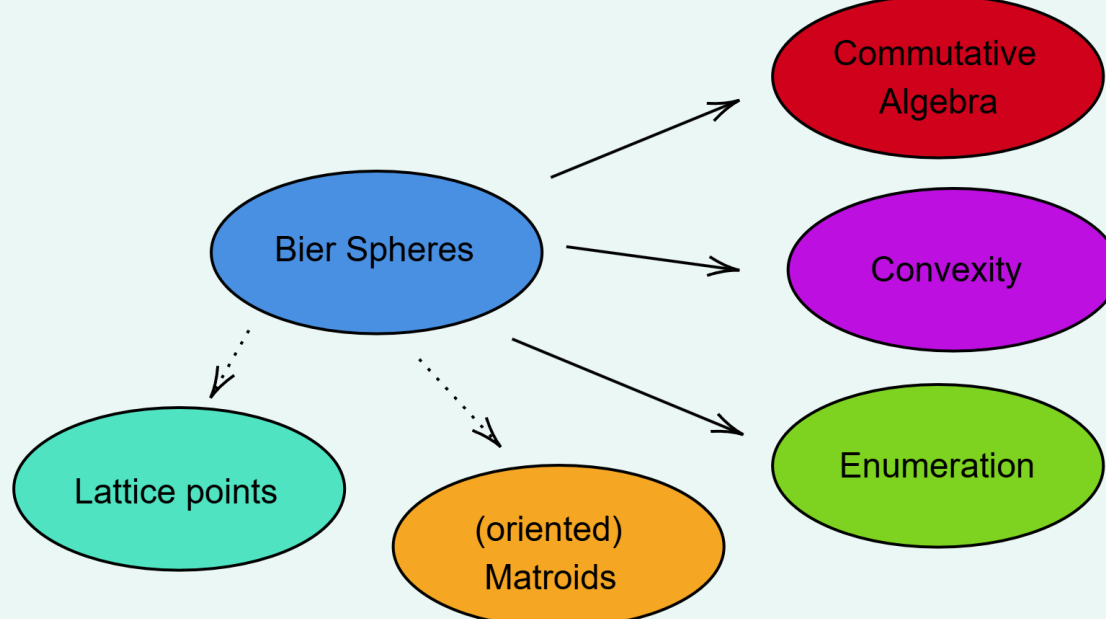
The combinatorial story: when is the independence complex of a graph unimodal?

- Conjecture.** (Alavi, Malde, Schwenk and Erdős, 1987) The independence polynomial of a tree is unimodal.
- The conjecture is still open, and it led to several generalizations (many of which are now known to be false). A special case that is still open is the following:
- Conjecture.** (Levit and Mandrescu, 2006) Let Γ be a $(d - 1)$ -dimensional pure flag complex on $2d$ vertices. Then the f -vector of Γ is unimodal.
- A special case of this conjecture is when Δ is a flag complex and $\Gamma = B_n(\Delta)$. After translating, this is actually the case that has attracted the most attention in graph theory.

Untold stories: Open questions & Further directions

1. The Bier construction has been generalized in several directions. Some lead to more triangulations of spheres/balls, and some lead to the triangulations of more complicated manifolds.
2. Can the algebraic/topological properties of Bier spheres be used to prove/disprove the Levit-Mandrescu conjecture?
3. Some Bier spheres are known to be Kantorovich-Rubinstein polytopes. This same class of polytopes is also related to linear sections of symmetric edge polytopes.
4. Can the partial chirotope defined by the orientation of a Bier sphere always be extended to a full chirotope?

Conclusion



Check out the algebra paper!



Check out the geometry paper!



Check out the paper connecting some of these ideas!